Inverse Trigonometric Functions and Their Derivatives

First Some Properties of Inverse Trig Functions . . .

Trig Table

$\theta°$	0°	30°	45°	60°	90°
$\sin\theta$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
$\cos\theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
$\tan\theta$	0	$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$	undefined

And now for the derivative of $y = \sin^{-1}(x)$

Find $\frac{dy}{dx}$: $y = \sin^{-1}(4x)$

Find $\frac{dy}{dx}$: $y = \sin^{-1}(x^3)$

Hmmm..what is the derivative of $y = \cos^{-1}(x)$?

Find $\frac{dy}{dx}$: $y = \cos^{-1}(x^5)$

Find $\frac{dy}{dx}$: $y = \cos^{-1}\left(\frac{1}{x}\right)$

In general....

$$\frac{d}{dx}\left[\sin^{-1} x\right] = \frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx}\left[\sin^{-1} u\right] = \frac{1}{\sqrt{1-u^2}} \cdot u'$$

$$\frac{d}{dx}\left[\cos^{-1} x\right] = -\frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx}\left[\cos^{-1} u\right] = -\frac{1}{\sqrt{1-u^2}} \cdot u'$$

And now for $y = \tan^{-1}(x)$ and $y = \cot^{-1}(x)$

Find $\frac{dy}{dx}$: $y = \tan^{-1}(3x^3)$

Find $\frac{dy}{dx}$: $y = \cot^{-1}(\sqrt{x})$

The derivative of $y = \sec^{-1}(x)$ and $y = \csc^{-1}(x)$

$$\frac{d}{dx}\left[\sec^{-1} x\right] = \frac{1}{|x|\sqrt{x^2 - 1}}$$

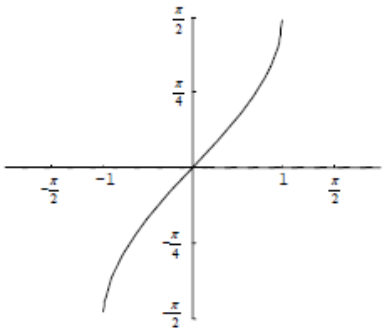
$$\frac{d}{dx}\left[\csc^{-1} x\right] = -\frac{1}{|x|\sqrt{x^2 - 1}}$$

The derivative of $y = \sec^{-1}(u)$ and $y = \csc^{-1}(u)$

Find $\frac{dy}{dx}$: $y = \csc^{-1}(x^2)$

Domain: $[-1,1]$

Range: $\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$

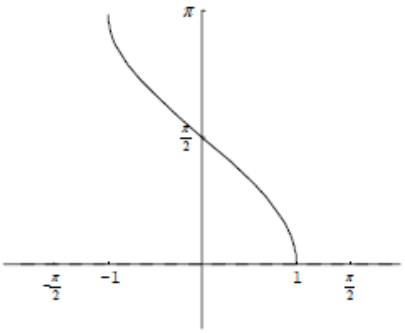


$f(x)=\sin^{-1}x$

$f(x)=\arcsin x$

Domain: $[-1,1]$

Range: $[0,\pi]$



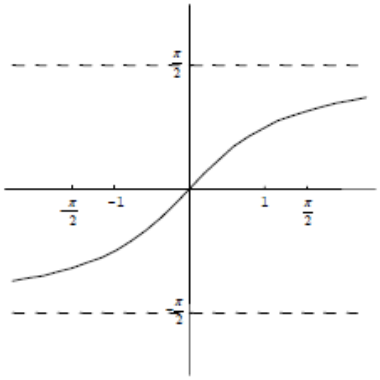
$f(x)=\cos^{-1}x$

$f(x)=\arccos x$

http://www.math.wisc.edu/~ysu/math221f12/graphsoftrigfunctions.pdf

Domain: $(-\infty,\infty)$

Range: $\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$

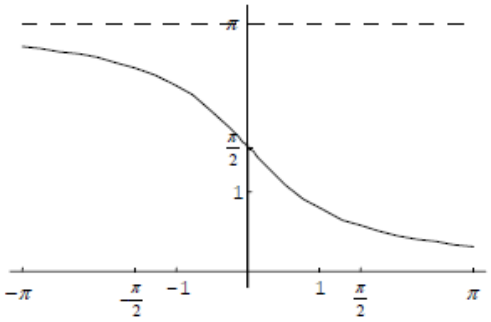


$f(x)=\tan^{-1}x$

$f(x)=\arctan x$

Domain: $(-\infty,\infty)$

Range: $(0,\pi)$

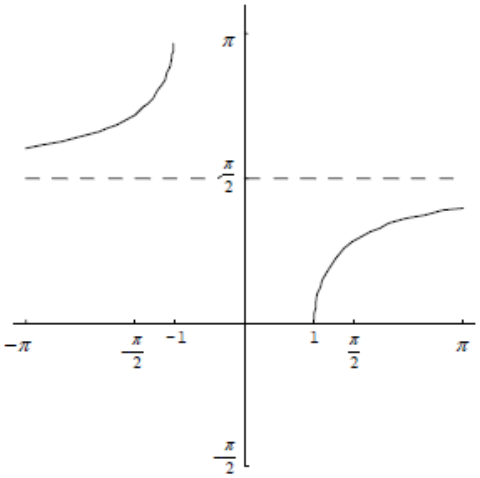


$f(x)=\cot^{-1}x$

$f(x)=\operatorname{arc}\cot x$

Domain: $(-\infty,-1]\cup[1,\infty)$

Range: $[0,\pi], y \neq \frac{\pi}{2}$

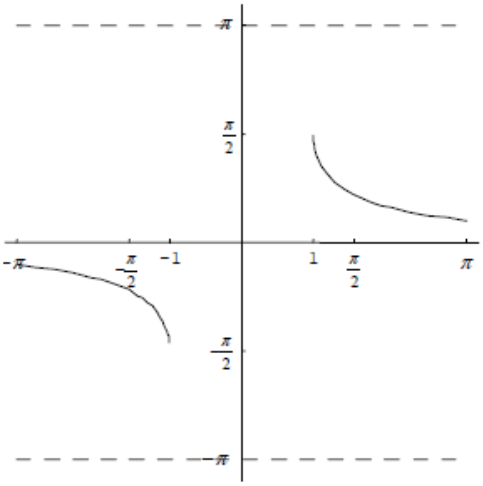


$f(x) = \sec^{-1} x$

$f(x) = \operatorname{arc}\sec x$

Domain: $(-\infty,-1]\cup[1,\infty)$

Range: $\left[-\frac{\pi}{2},\frac{\pi}{2}\right], y \neq 0$



$f(x) = \csc^{-1} x$

$f(x) = \operatorname{arc}\csc x$